VAR Pipeline — Plain-English Results Report

Media Bias × Stock Returns Study | S&P 500 Panel | 2015–2025

Steps 1–8 completed — produced May 2026

1. What This Study Is Doing

This study asks a simple question: if news coverage about a company becomes more negative (or positive), does the company's stock price tend to move in the following weeks? To answer it, we built a pipeline that goes from raw news articles all the way to a statistical test called a VAR (Vector Autoregression).

We also have a second question layered on top: did any relationship between news and stock prices change after the COVID-19 shock in 2020? This is answered separately using a different method called Difference-in-Differences (DiD). That part (RQ1 and RQ2) has not been run yet — this report covers the VAR pipeline only.

> **The five research questions**
>
> RQ1: Did media coverage (bias) toward S&P 500 firms change after the 2020 recession? (DiD — not yet run)
>
> RQ2: Did stock returns change after the 2020 recession? (DiD — not yet run)
>
> RQ3: Does media bias predict stock returns at all? (VAR — DONE, results in this report)
>
> RQ4: Did the link between bias and returns strengthen post-2020? (DiD + VAR subsample — partially done)
>
> RQ5: Does bias affect Indian markets more than US markets? (future work)

2. What a VAR Is — In Plain English

A VAR (Vector Autoregression) is a way of asking: "does variable A from last week help predict variable B this week, even after accounting for what B was doing last week on its own?" It tests both directions at once — does bias predict returns, and do returns predict bias?

The specific test we use inside a VAR is called a Granger causality test. "Granger causality" does not mean real causality — it means predictive precedence. Formally: "does knowing past values of X make my forecast of Y more accurate than using past Y alone?" If yes, X Granger-causes Y.

> **What "passing" looks like for each test in this pipeline**
>
> Stationarity (ADF + KPSS): the series should not drift or trend over time. Pass = confirmed stationary.
>
> VAR stability: all mathematical roots of the model must be > 1. Pass = model is not exploding.
>
> Portmanteau test: residuals should not be autocorrelated. Pass = $p > 0.05$.
>
> ARCH test: checks for volatility clustering. No strict pass/fail — just tells us to use robust standard errors.
>
> Granger causality: the key test. Pass (significant) = $p < 0.05$ means X predicts Y.

3. Step 1 — Data Inventory

What we did

Checked how many articles we have per company and how consistently they appear over time. Also checked the stock price data for gaps or mismatches.

What we found

Finding	Detail
26 companies in article data	8 were too sparse to model — almost no articles for most days.
8 companies dropped	MMM, AAPL, TSLA, NKE, PGR, ORCL: sparse articles (likely data artefact, not real). ETN and NDAQ: essentially zero coverage (5 and 43 articles over 10 years).
4 in stock data but no articles	NVDA, PFE, TGT, VZ — never scraped or out of scope. Excluded.
BAC and FDX: truncated stock data	Bank of America stock data only starts Dec 2020. FedEx starts Apr 2021. Cannot be used for any pre/post comparison.
V and WMT: data bug	Visa and Walmart have identical stock returns for all 522 weeks — same price series was assigned to both. WMT excluded until fixed.

Changes to the plan

- Dropped 10 companies from the VAR. Final sample: 15 companies.
- Confirmed weekly frequency is correct — daily has too many zero-article days for most firms.
- The "daily" recommendation from the automated script was wrong — overruled based on distribution inspection.

4. Step 2 — Bias Index Construction

What we did

Built the weekly bias index B(i,t) for each company. Each article gets a stance score between -1 and +1 (from the NewsMTSC model). We averaged these per company per week.

What we found

Finding	Detail
No out-of-range scores	All article scores fall within [-1, +1]. Pipeline is working correctly.
Negative skew across almost all companies	Financial journalism tends to frame negative events more dramatically. Normal finding, not a bug.
WFC and UBER persistently negative	Wells Fargo mean = -0.28, Uber mean = -0.24. Real — both had sustained scandals/regulatory issues. Flag for stationarity testing.
Netflix: dropped	234 of 574 weeks have zero articles including entire years (2020, 2021, 2023 completely missing). Cannot model. Dropped.
Simple mean chosen over weighted mean	Weighted mean diverges significantly for UBER, GM, SBUX. Simple mean is more robust.

Changes to the plan

- Netflix dropped from all VAR analysis.
- Use simple_mean_stance as the bias index throughout.
- 16 companies remain after this step.

5. Step 3 — Gap Analysis & Imputation

What we did

Looked for weeks with no articles (gaps in the bias series) and filled them in using simple rules. Gaps of 1-2 weeks: carry the last known value forward. Gaps of 3-8 weeks: interpolate linearly between surrounding values. Gaps longer than 8 weeks: do not fill — exclude that period.

What we found

Result: almost nothing to fix

Only 58 weeks were imputed across 16 companies × 574 weeks = 9,184 total observations. That is 0.6% imputed. Negligible.

Finding	Detail
All 16 companies pass coverage threshold	Every company has >97% weekly coverage. No company dropped here.
Longest gap anywhere: 7 weeks	Occurs in T, SBUX, UBER, V, WMT, WFC — all at the very end of the sample (Nov-Dec 2025). End-of-sample, low impact.
Linear interpolation mislabelled for end-of-sample gaps	When the gap is at the end of the series there is no future value to interpolate toward, so the method defaults to forward-fill. Functionally correct, just mislabelled.
WFC row imputed at -0.621	Extreme but correct — the source week genuinely had that score. Not a bug.

6. Step 4 — Returns Construction & Alignment

What we did

Computed weekly log returns from adjusted daily closing prices: sum of ln(P_today / P_yesterday) across all trading days in the week. Then merged the returns with the bias index to get one aligned dataset per company.

What we found

Finding	Detail
All 16 companies have ≥100 aligned weeks	Minimum needed for VAR. All pass. Most have 522 weeks.
AT&T has 18 extreme return weeks (>15%)	All confirmed as real events: earnings shocks, COVID crash, debt restructuring. Do not winsorise. AT&T is genuinely volatile due to its debt load.
V and WMT identical returns confirmed	Bug verified — all 522 return values are identical. WMT excluded.
BAC aligned window: 260 weeks	Post-2020 only. Fine for full-sample VAR but cannot contribute to pre/post comparison.

7. Step 5 — Stationarity Testing

What this test is

A time series is "stationary" if it does not trend or drift over time — it bounces around a stable average. VAR requires stationary inputs. We used two tests:

- ADF test — null hypothesis is "the series has a unit root (non-stationary)". Rejecting it ($p < 0.05$) means stationary.
- KPSS test — null hypothesis is "the series is stationary". Rejecting it ($p < 0.05$) means non-stationary.
- If both agree: clear result. If both reject: structural break likely (the mean itself shifted at some point).

What we found

Series	Result
Weekly log returns	Stationary (I(0)) for all 16 companies. Expected — returns always are.
Bias index at levels	"Conflicting" for 14 of 16 companies — both tests reject simultaneously.
What "conflicting" actually means	Not a problem. The 2020 shock caused a level shift in media sentiment. ADF sees mean-reversion within each period; KPSS sees the shift across periods. Both are right. The series is I(0) with a structural break.
Bias index first-differenced (ΔBias)	Stationary for all 16 companies. Clean pass.
WFC special case	Cointegration test flagged. Likely also due to structural break. Treated same as others — VAR in levels with ΔBias.

Decision made

VAR uses ΔBias (first-differenced) and Returns (levels)

Taking first differences of the bias index removes the drift and structural break issues. This means the VAR is testing whether changes in media sentiment predict returns, not the level of sentiment. This is actually a cleaner economic claim: a sudden shift in tone is more likely to move markets than a persistently negative baseline that investors have already priced in.

8. Step 6 — Structural Break Testing

What this test is

The Bai-Perron test finds the single largest shift in the mean of a time series. We used it to find when media sentiment structurally changed for each company — to define the "before" and "after" periods for RQ3.

What we expected vs what we found

Surprise: COVID-19 (Jan 2020) is NOT the dominant break for most companies

We expected most bias breaks to cluster around early 2020. Instead, most fall in 2022-2024. Only Visa (Sep 2020) is close. This means the media sentiment shift happened later than the COVID onset for most firms.

What likely drove the 2022 cluster: the post-COVID macro regime change — interest rate hikes, inflation coverage, end of the zero-rate era. These structural shifts in the economic narrative drove lasting changes in how firms were covered.

Company	Bias Break Date	Interpretation
AMZN, UBER	2018	Pre-COVID — antitrust scrutiny (AMZN), safety scandals (UBER)
GS, WFC	2019	Pre-COVID — regulatory/financial sector events
V	Sep 2020	Closest to COVID — payment network uncertainty
SBUX, BAC, T, MCD, GM	2021-2022	2022 macro regime shift cluster
MSFT, MS, INTC	2023-2024	AI narrative shift — coverage restructured around AI winners/losers
BA, WMT	2024	Late breaks — Boeing crisis coverage, Walmart e-commerce narrative

Impact on the pipeline

- Each company gets its own firm-specific Post(t) dummy based on its empirical break date, not a universal Jan 2020 cutoff.
- For RQ1 and RQ2 DiD regressions (not yet run): keep Jan 2020 as the theoretical anchor, report Bai-Perron as a robustness check.

- Four companies have thin post-break windows: BA (99 obs), INTC (88), WMT (71), MS (103). Results for these are indicative only.

9. Step 7 — VAR Lag Selection & Diagnostics

What this step does

A VAR needs to know how many past weeks to look back (the "lag order" p). We tested p = 1 through 8 for each company and picked the value that balances fit vs. complexity using the BIC criterion.

What we found

Diagnostic	Result
Stability	All 15 models stable. All minimum eigenvalue roots > 1. Pass across the board.
BIC lag selection	Most companies chose p = 3. Modal lag = 3, used for panel VAR.
AIC vs BIC gap	AIC wants 6-8 lags everywhere; BIC selects 2-5. Wide gap. Sensitivity check needed.
Portmanteau test (autocorrelation)	11 of 15 companies fail ($p < 0.05$). Means residual autocorrelation remains — the lag order may be slightly too low.
Jarque-Bera (normality)	All 15 fail. Expected for financial data — fat tails everywhere. VAR estimates still valid.
ARCH test (volatility clustering)	10 of 15 companies have significant ARCH in returns. Common for stocks.

Decision made

> **Use HAC-robust standard errors throughout**
>
> Because of the autocorrelation and ARCH issues, all Granger tests use Newey-West HAC-robust standard errors. This is standard practice for financial time series and corrects for both problems simultaneously without needing to add more lags.

10. Step 8 — Granger Causality Results

What changed from the original plan

Originally the plan was to run 15 separate firm-level VARs and test Granger causality for each company. Midway through, the research question was reframed to ask whether media bias predicts returns generally across the market — not just for individual firms. This required a Panel VAR (pooling all firms together) as the primary result, with firm-level results kept as a secondary appendix.

Panel VAR — the main result (RQ5)

Panel Granger result: CLEAR NULL

Bias → Returns: $p = 0.976$ (full sample), $p = 0.245$ (pre-break), $p = 0.617$ (post-break)

Returns → Bias: $p = 0.200$ (full sample), $p = 0.801$ (pre-break), $p = 0.187$ (post-break)

Across 7,493 pooled observations with VIX, S&P 500 returns, article count, and firm fixed effects all controlled — media bias does NOT predict stock returns at the panel level. The reverse channel also does not hold.

What this finding means

This is not a failure — it is a finding. For large, liquid, heavily-covered S&P 500 firms, media sentiment at weekly frequency has no predictive content for next week's returns once market-wide factors are controlled for. This is consistent with market efficiency: for these firms, information in news articles is already priced in quickly, before the week is out.

Firm-level results — heterogeneity appendix

The firm-level tests (run separately per company) show some variation:

Company	Bias → Returns	Returns → Bias
WFC (Wells Fargo)	p = 0.023 ✔ SIGNIFICANT	p = 0.307 (none)
MS (Morgan Stanley)	p = 0.058 ≈ borderline	p = 0.016 post-break ✔
GS (Goldman Sachs)	p = 0.757 (none)	p = 0.012 ✔ SIGNIFICANT
MCD (McDonald's)	p = 0.077 post-break ≈	p = 0.027 pre-break ✔
All others (11 firms)	Not significant	Not significant

WFC is the strongest individual result: bias predicts returns (p=0.023 full sample, p=0.013 post-break) but not pre-break. This fits the narrative — Wells Fargo's sustained negative regulatory coverage post-2019 had genuine predictive content for its stock. GS shows the reverse: price moves predict subsequent media coverage, not the other way around.

11. What These Results Say About Your Research Questions

RQ3 — Does media bias predict stock returns?

Answer: No, at the panel level. Maybe, for specific firms.

At the pooled S&P 500 level: no. Media bias does not generally predict weekly returns for large-cap US stocks.

At the firm level: Wells Fargo shows significant Granger causality. Morgan Stanley shows borderline evidence.

The honest claim: media sentiment has predictive content for returns in firms with persistent, high-salience negative narratives (regulatory scandals, sustained reputational damage). It does not work as a general market signal.

RQ4 — Did the bias-return relationship strengthen post-2020?

Answer: Partial evidence, firm-specific.

For WFC: pre-break $p = 0.345$ (nothing), post-break $p = 0.013$ (significant). The relationship emerged post-break. This is exactly the RQ3 pattern.

At the panel level: no structural change detected (pre $p = 0.245$, post $p = 0.617$).

Conclusion: the post-2020 strengthening happened for specific firms, not the broad market. RQ3 will be better answered once the DiD regressions (RQ1 + RQ2) are run — the VAR subsample results provide corroborating evidence rather than the primary answer.

RQ1 & RQ2 — Did bias and returns shift post-2020?

Not yet answered.

These require DiD regressions which have not been run yet.

The structural break testing (Step 6) has confirmed that break dates vary by firm — most are in 2022, not 2020.

Recommendation: run the DiD with Jan 2020 as the theoretical cutoff, then show the Bai-Perron firm-specific dates as robustness.

Note: RQ3 is conditional on RQ1 and RQ2 both being significant. If either is null, RQ3 needs reframing.

12. What Is Still To Do

Task	Notes
Fix V / WMT data bug	One of these has the wrong price series. Recheck the stock data source for both tickers and re-run Steps 4 onward for the corrected company.
Run DiD for RQ1 and RQ2	Panel OLS with firm fixed effects, Post(t) dummy, VIX and market return controls. Run on the bias series and returns series separately.

Task	Notes
Run RQ4 triple-interaction DiD	Only after RQ1 and RQ2 are confirmed significant. Requires a treatment/control group definition.
Step 9: IRF and FEVD	Run impulse response functions for WFC and MS only (the two firms with meaningful Granger results). Panel IRF is not meaningful given the null panel result.
Robustness checks	Re-run panel Granger at AIC lags (6-7) to confirm null result holds. Re-run on raw (non-imputed) series.
Indian market data (RQ4)	NIFTY 50 article scraping and stock data not yet integrated. Future work.
Write methodology and results sections	The VAR pipeline is complete. All inputs for the methods section and results section are now available.

13. Quick Reference: Final Company Status

Company	Status	Notes
AMZN, T, BA, GM, GS, INTC, MCD, MSFT, MS, SBUX, UBER, V, ABNB	In VAR ✔	Full pipeline completed. Null Granger result.
WFC	In VAR ✔ — key finding	Significant bias → returns post-break. Headline firm result.
BAC	In VAR ✔ — post-2020 only	Stock data only from Dec 2020. Cannot contribute to pre/post comparison.
WMT	Excluded — data bug	Identical returns to Visa. Re-include after fixing price source.
NFLX	Excluded — Type C gaps	Entire years of missing article data. Cannot model.
MMM, AAPL, TSLA, NKE, PGR, ORCL	Excluded — sparse articles	Likely scraping artefact. Note as limitation in paper.
ETN, NDAQ	Excluded — near-zero coverage	5 and 43 articles over 10 years. Footnote only.
NVDA, PFE, TGT, VZ	Excluded — no article data	In stock dataset but never scraped. Document and exclude.

End of report